POLYPOD: an Artificial Reef for the Coral Gardens of the Future

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Abstract
Coral reefs sustain a quarter of marine biodiversity and shield coastlines from up to 97% of incoming wave energy, yet 84% of the world's reef area was bleached in the 2023–2025 event and the IPCC projects that 70–90% of modern reefs may be lost at 1.5 ℃ of warming. Existing artificial-reef instruments fall on opposite sides of an unresolved trade-off: opportunistic deployments (tyres, debris, sunken vessels) are cheap but ecologically harmful, while contemporary engineered prototypes are biologically and hydrodynamically excellent but priced out of reach for the rural coastal communities that need them most. We present POLYPOD, a modular, ceramic, slip-cast artificial reef unit designed around a single competing objective: state-of-the-art biological and engineering performance at a cost and deployment profile accessible to individual actors. A single 3D-printed master pattern produces a reusable plaster mould from which ceramic units can be cast at ∼$2–16 per piece depending on scale and assembled by a single diver. Computational fluid dynamics validation shows the structure dissipates approximately 60% of incoming wave energy. The aspiration is to make POLYPOD the substrate of community-scale coral gardens: objects that an individual can buy, transport without machinery, and deploy together with a nursery-grown fragment, so that creating a positive impact on the ocean becomes as accessible as causing harm to it.

Keywords
Ocean Tech — Habitat Restoration — Sustainable Design — Modular Design

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Introduction

Coral reefs occupy less than 0.1% of the ocean's surface, yet they sustain roughly 25% of all known marine species, making them the most biodiverse marine ecosystems on Earth [1]. This ecological productivity translates directly into economic value: coral reefs generate an estimated $3.6 billion in net annual benefits in the US alone through tourism, fisheries, and coastal protection, supporting millions of jobs in coastal and island communities [2, 3]. Healthy reefs can absorb up to 97% of incoming wave energy [3], making them critical coastal infrastructure. Their loss simultaneously threatens biodiversity, local food security, and the physical protection of coastlines, risks that fall disproportionately on rural coastal communities whose livelihoods depend on small-scale fisheries and reef-based tourism [4, 5].

Despite their importance, coral reefs are undergoing an accelerating crisis. The IPCC has estimated that 70–90% of modern coral reefs may be lost if global temperatures rise to 1.5°C above preindustrial levels, rising to more than 99% at a 2°C increase [6]. The 2023–2025 bleaching event, now confirmed as the fourth global coral bleaching event on record, affected 84% of the world's reef area across at least 82 countries , with coral mortality reaching 93% in some Pacific regions [7]. The IPCC Sixth Assessment Report found that the frequency and severity of mass bleaching events have increased sharply since the early 1980s [8], and modelling work indicates that even the most optimistic mitigation pathways project substantial reef degradation across all major tropical marine regions by century's end [9].

Two complementary palliative strategies have been widely adopted in response: artificial reef deployment and coral gardening. Artificial reefs provide stable, chemically compatible substrate for coral regrowth, shelter for juvenile fish, and coastal defence in areas where natural reefs have been severely degraded or where the ocean floor cannot host coral aggregation. Coral gardening, in parallel, propagates coral fragments harvested from donor colonies or as "corals of opportunity,"

grows them to viable size in nurseries, and outplants them onto degraded substrate [10]. The two approaches are mutually reinforcing: the substrate onto which gardened corals are outplanted is the single largest determinant of transplant success [11]. Recent demonstrations show that corals can now be grown at scale and at very low cost, with survival rates exceeding 94% for *Acropora* species transplanted at just USD 2.50 per coral [12]. This opens a possibility that was not viable a decade ago: community-based local restoration in which individuals, rather than governments or large NGOs, are the first actors.

The artificial reefs currently available, however, offer trade-offs that are poorly matched to this vision. Existing instruments vary significantly in price, functionality, coastal protection capability, material safety, and deployability [13, 14, 15], and none simultaneously satisfies the constraints of a community-scale coral garden.

In this sense, this work proposes a synthesis of ecological performance and economic viability, competing but jointly necessary objectives. We introduce POLYPOD, a modular, ceramic slip-cast artificial coral reef unit that adheres to state-of-the-art biological and engineering standards while targeting a cost and deployment profile accessible to individuals and coastal communities. As depicted in our storyboard (Fig. 1), the aspiration is for it to become the substrate for future community coral gardens: bought by members of the community, transported without heavy machinery, placed by a single diver, and paired with corals grown from a nursery. We hope to leverage individual enterprise in this fight, creating a world in which it is as easy for an individual to create a positive impact on the ocean as it is to cause harm. The instrument also carries entertainment, business and educational potential, fostering tourism and growing a generation protective of marine ecosystems.

1. Related Work

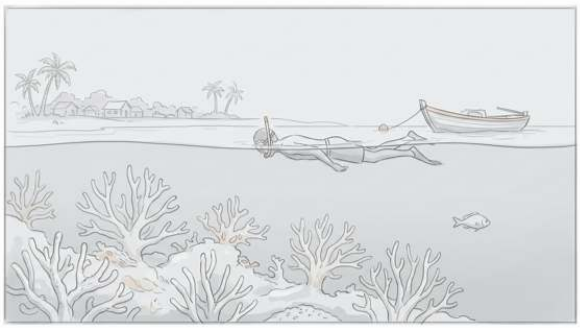
Biological case Coral restoration operates on a well established biological principle: corals require a stable, chemically compatible, and texturally complex substrate on which their larvae can settle and metamorphose into sessile polyps [13]. When degraded natural substrate is unavailable, artificial reefs (ARs), human-created structures designed to promote marine life and, in coastal applications, attenuate wave energy, can provide this foundation. A systematic review by Ref. [16] of 136 AR deployments in tropical and subtropical coral reefs found that the most commonly reported outcomes were increases in fish abundance (55% of studies), enhancements in habitat quantity (31%), and gains in coral cover (27%). Crucially, success rates were highest for the provision of nursery habitat and increases in coral cover, each achieving their stated objectives in approximately 71% of cases [16]. A parallel meta-analysis by Ref. [17] corroborated these findings, identifying coral transplantation, larval resettlement, and habitat mitigation as the most productive applications of artificial reef technology [17].

Coral gardening complements substrate-based approaches and is increasingly mainstream. Nurseries operate in two configurations: field-based (*in situ*) nurseries suspend fragments on underwater rope or net structures, while land-based (*ex situ*) facilities offer environmental control and selective breeding of thermally tolerant genotypes. A global review covering more than 360 restoration projects across 56 countries found that 48% involved coral gardening in some form, with average post-transplantation survival of 66% and 20% of projects exceeding 90% survival [10]. The intersection of coral gardening with AR design is intriguing: in absence of a favorable seabed or of enough degraded coral, nursery-grown corals are outplanted onto artificial substrate. The surface chemistry, texture, and morphology of the receiving artificial substrate become decisive factors in transplant success [11].

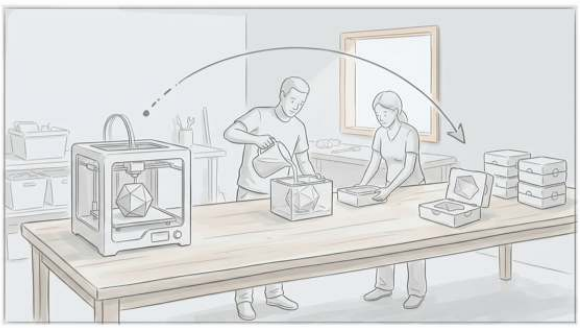
Artificial Reefs Existing AR instruments fall into three broad categories. *Early-generation deployments* such as scrap tyres, decommissioned vessels, and construction debris introduced persistent pollutants. Plastic particles in particular have been documented to trigger coral pathologies and high polyp mortality [13] and are therefore inadapt as a receving substrate. *Contemporary research prototypes*, such as the MARS / MARS II structures from Reef Design Lab, represent genuine scientific and design advances but at unit costs inaccessible to the rural coastal communities that most need them [15]. Another example is Archireef (HKU spin-off, Hong Kong), that produces terracotta tiles that are easy to assemble and showcase high coral survival rates, but are designed for restoration only and provide no coastal-defence function [18]. Finally, *promising academic systems* such as the University of Miami's SEAHIVE / X-REEFS [14], demonstrate excellent wave dissipation and ecological integration in controlled settings but do not allow for scalable and accessible deployment. None of these instruments simultaneously satisfies low cost, non-toxic substrate, single-diver deployability, and meaningful wave attenuation: the configuration we aim to for a community-scale coral garden.

2. Methods

Modular design We design *POLYPOD* as a modular $40 \times 40 \times 25$ cm double v-shaped unit so that a single mould can serve variable site conditions. Its design is inspired by coastal breakwater structures called stabits. Here modularity serves three project-defining goals simultaneously. (i) It compresses the manufacturing complexity onto a single mould, which is what makes the per-unit cost and the kit-distribution model viable in the first place. (ii) It keeps every individual piece light and small enough for one-person handling, which is what makes single-diver deployment possible. (iii) It allows local communities to tune the configuration to local water depth, target species, and the level of wave attenuation required, without ever needing custom tooling. Two basic ensembling strategies are shown in Fig. 2 and 3), but different configurations are achievable. The interlocking mechanism is inspired



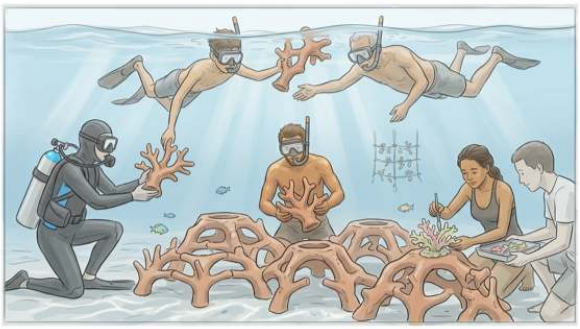
Seeing the loss A villager witnesses the silence of a dead reef and realizes the urgent need for restoration.



From digital to physical A simple local workflow turns digital reef designs into plaster molds for community use.



Casting at home Families cast reef units at home, making restoration part of everyday life.



Deployed hand-in-hand Community members work together in the water to assemble the reef and attach young corals.

Figure 1. Our storyboard. From the realization of the coral reef loss to the personal initiative of communities, enabled by our modular, eco-compatible and economically viable design and by current coral gardening techniques

by interlocking bricks and yet to be refined. It currently comprises of vertical and lateral interlocking mechanism and the possibility to add vertical poles on the sides to increase the overall stiffness of the structure. Future simulation or experimental work will be performed to further validate the best interlocking design.



Figure 2. Horizontal configuration of POLYPOD

Material selection The history of AR material reflects a progression from opportunistic reuse toward purpose-engineered substrates. Conventional concrete, while structurally durable, is chemically incompatible with coral settlement: once submerged, its surface pH of 10–11 suppresses larval settlement for months, during which pH-resistant barnacles colonise the surface and competitively exclude corals [13]. Plastic-composite materials introduce a further hazard: plastic particles release phagostimulants that render them indistinguishable from prey, disrupting coral feeding, clogging digestive systems, and facilitating pathogen introduction [13].

Against this background, ceramics are the leading substrate candidate. Ref. [13] synthesise the evidence: ceramics are chemically inert, non-toxic, and have shown high rates of coral larval settlement in controlled trials [13]. Ref. [19] demonstrated that porous ceramics with tailored pore sizes (400–800 μm, matched to larval body dimensions) produced measurable improvements in settlement success. Most earthenware also has the advantage of low-energy firing, an important consideration for cost reduction [20]. We have therefore selected unglazed ceramics as the material for our prototype but terracotta would also be an option as the fabrication technique is independent of the choice of the two.



Figure 3. Vertical Configuration of POLYPOD

Biomimicking Beyond material chemistry, surface morphology is a primary determinant of larval settlement, and the literature converges on a multi-scale texture framework spanning three hierarchical levels [19, 21, 22, 23]. At the **macro scale** (10–50 mm), caves, overhangs, and channels provide shelter for fish. At the **meso scale** (1–5 mm), millimetre ridges extend the larval settling window by over 300% and yield settlement rates an order of magnitude higher than on flat surfaces [22]. At the **micro scale** (400–800 μm), pore sizes matched to larval body dimensions act as micro-refugia, protecting newly settled larvae from predation and mechanical disturbance [19]. The combined effect is well-documented: Archireef has reported 95% coral survival on algorithm-generated terracotta tiles mimicking brain-coral surface patterns, roughly four times the survival rate of conventional transplanting methods [18].

POLYPOD addresses the macro and meso scales directly through cast geometry, which we generated algorithmically to produce coral-like patterns (Fig. 5). The micro-scale target is harder to satisfy under our cost constraints. Reliable, repeatable porosity in the 400–800 μm range is achievable in principle but typically requires specialised equipment that we are unwilling to push onto the end user without compromising the project's core value proposition of decentralised, low-cost fabrication. We therefore rely on the natural surface porosity of unglazed terracotta as a first-order approximation, and defer the development of a kit-friendly micro-porosity protocol to future work. Fig. 4 and Fig. 5 illustrate how the current design realises the upper two scales of the framework.

Fabrication technique We fabricate POLYPOD by slip casting, a low-energy ceramic process in which liquid clay is poured into an absorbent plaster mould; the mould draws water from the slip, depositing a uniform clay layer against its walls before the excess is poured out. Slip casting is the right fit for this project for three reasons. First, a single mould shipped with the final kit produces a virtually unlimited number of identical casts, amortising shipping and tooling costs across the entire production run. Second, the process requires no specialised machinery beyond the mould itself and can be performed and repeated easily by individuals provided with good instructions, enabling decentralised production close to deployment sites. Third, the workflow naturally splits into steps that match the kit-distribution model: the upstream tasks (3D-modelling and printing the master geometry, casting the plaster mould) are performed once at a centralised facility, while the downstream tasks (slip preparation, casting, demoulding, and, potentially, firing) are simple enough to be performed by community members. Our pipeline consists of (i) 3D-printing the master mould pattern (Fig. 6), (ii) casting the plaster working mould around it (Fig. 7), and (iii) slip-casting the ceramic unit (a picture of the prototype is shown in Fig. 8). Further considerations regarding firing are addressed in the Limitations section. A step by step guide of the fabrication is shown in the video.

Coastal-defence CFD validation We validate the coastal-defence performance of our artificial reef through computational fluid dynamics, computing the wave power dissipation when passing through the structure. We simulate a single laminar wave in a 12m long 1 m wide domain. The structure is approximately 1 m tall, 80 cm in the other dimensions (Fig. 9 and **??**). The instantaneous power-flux density carried by a propagating wave is

$$S_x = p\,u_x, \tag{1}$$

where p is the dynamic pressure and $\mathbf{u}$ the fluid-velocity vector, and u_x is its component perpendicular to the structure (Fig. 10). The total instantaneous power crossing a vertical control surface A normal to the wave is obtained simply by integrating:

$$P(t) = \iint_A p(\mathbf{x},t)\,u_x(\mathbf{x},t)\,\mathrm{d}A. \tag{2}$$

We define two such control surfaces: A_{in}, placed upstream of the reef, and A_{out}, placed downstream. We showcase how the total power across the surfaces evolves in time in Fig. 11. The presence of the artificial reef yields a net diminishment of 60% in wave power.



Figure 9. Side view of the computational domain for the CFD simulation

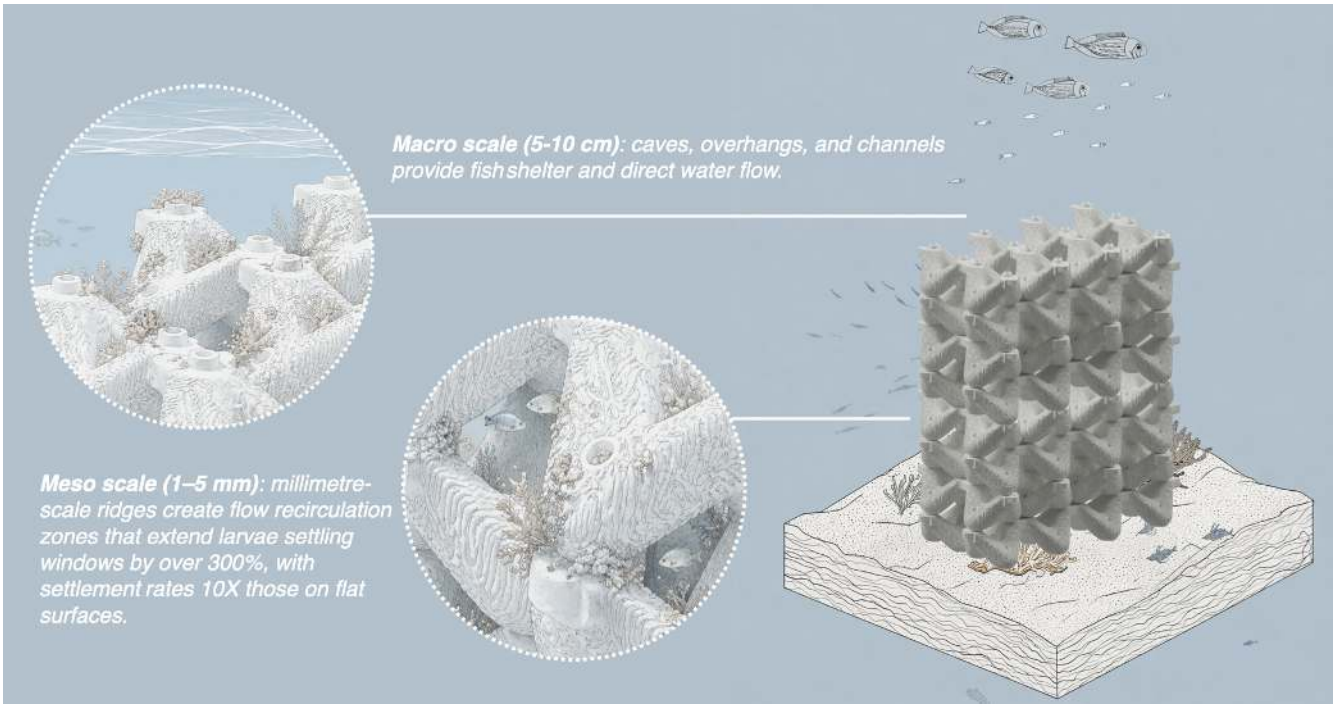


Figure 4. The design of POLYPOD adheres to the biologically validated and recognised standard of hierarchical multi-scale texture.



Figure 5. Three examples of ceramic surface patterns that biomimick natural coral patterns.

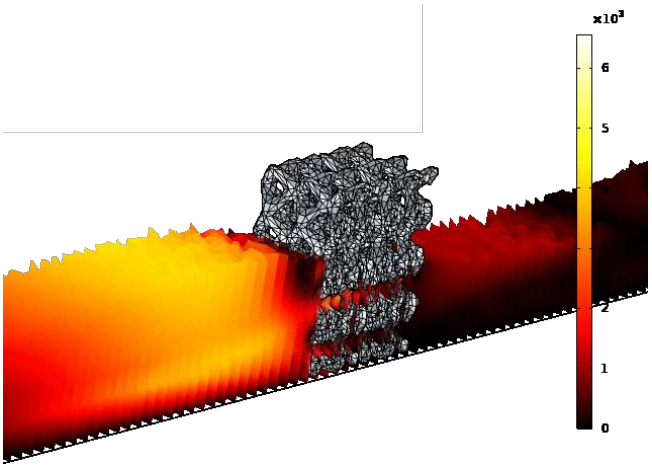


Figure 10. Depiction of the power density at time $t = 1$s. The lighter the more powerful is the wave at that point. We can clearly see how the power is dissipated passing through the structure.

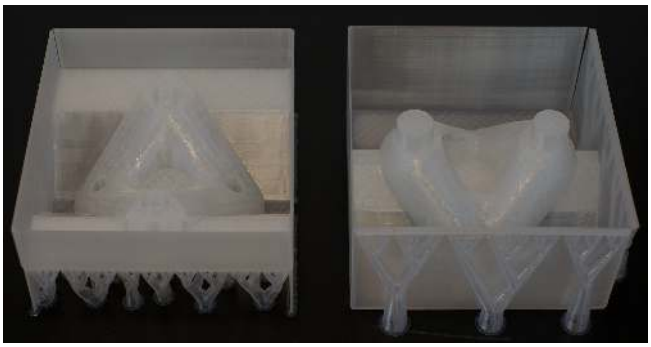


Figure 6. Two-part 3D printed mold of the mold. This is the *negative of the negative* with respect to our initial master. No requirements on the filament. The 3D printed shape must be bendable and with draft angles to allow for easy extraction.

Economic Feasibility We estimate fabrication cost from a tabletop run in which we slip-cast four prototype POLYPOD units at approximately one-third of the final intended size. The one-time inputs are the 3D-printed master pattern (∼1Kg of PLA filament at ∼\$10/kg, ≈\$10) and a single 1 kg (2.2 lb) batch of plaster of Paris used to cast the working mould (≈\$1.20 at bulk hardware-store pricing). The per-cast consumable is approximately 0.5 kg of ceramic slip per piece, at ≈\$3.50/kg, giving ≈\$1.75 of clay per unit. A single plaster mould of this size reliably yields 50 or more slip casts before surface degradation, so the one-time cost amortises rapidly. The first four pieces from a fresh mould therefore cost ≈\$20 in materials, or \$5 per piece; over 50 casts the per-piece cost

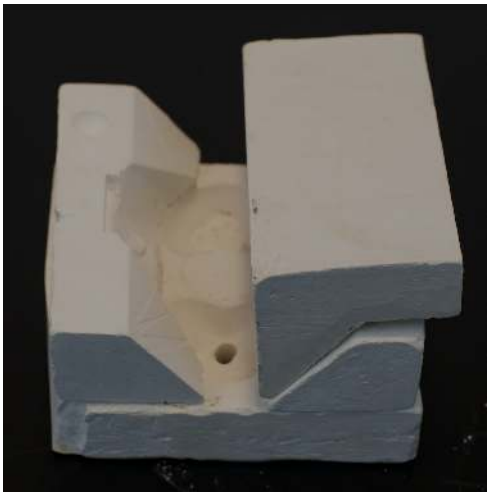


Figure 7. Dried plaster mold. This corresponds to an earlier version with a 5-part mold. After realizing there was a two part mold that yielded the same result without overhangs we moved towards that design.



Figure 8. On the left a casted POLYPOD prototype fully air-dried. On the right a prototype that was just slip casted and is still *leather-hard*.

reduces towards the slip cost alone. At full deployment scale, where the linear dimensions are roughly $3\times$ the prototype, material consumption per unit grows by a factor of $\sim 3^2 = 9$ because the slip-cast piece is hollow and slip is deposited only along the mould walls; wall thickness, set by absorption time, remains approximately constant across scales. Per-unit material cost therefore rises to roughly \$16–18, still one to two orders of magnitude below the per-unit cost of contemporary research prototypes [15]. This does not account for firing, another per-unit cost that is difficult to predict.

3. Limitations

A practical limitation of this workflow is that firing requires access to an electric kiln, which is not universally available. We considered alternatives, like air-dry clay or geopolymer cements, and rejected each: unfired and air-dried clay redisperses on prolonged contact with seawater, and the alkaline alternatives reintroduce the very pH-incompatibility problem that disqualifies conventional concrete. Firing is what converts the green slip-cast piece into a chemically stable, water-insoluble ceramic and is therefore non-negotiable for a substrate intended to remain submerged for years.

Firing may nonetheless seem counter-intuitive on environmental grounds, since electricity generation still releases greenhouse gases. We performed a calculation to bound this concern. A modern small electric kiln fires roughly 20 pieces per cycle at the low temperatures required for terracotta or other earthenware ($\sim$1000 °C), consuming on the order of 25 kWh. At the current US electric grid average carbon intensity [24], this corresponds to roughly 13 kg of CO_2 per firing, or **less than 1/27 of the emissions attributable to a single passenger on a 4-hour commercial flight for 20 pieces** ($\approx$360 kg CO_2, at 90 kg CO_2 per passenger-hour [25]). The figure improves further in countries where electrical grid rely more on cleaner sources of energy. Set against the road and air transport, the firing step is a negligible contributor to the overall carbon footprint in the world and of the project (assuming transportation carbon costs).

Acknowledgments

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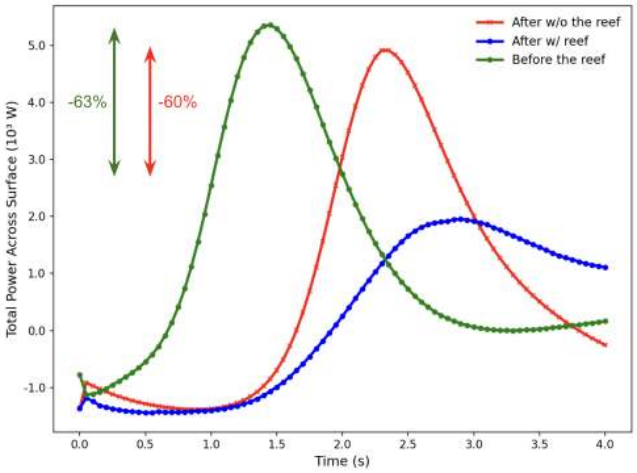


Figure 11. Time evolution of total power of a vertical surface. Here we are interested in seeing how the peak is dissipated by our structure. We can compare the peak value at the same vertical control surface with (blue) and without (red) the structure. There is a total dissipation of 63% and a net dissipation of 60% (i.e. not accounting for the natural dissipation of the wave).

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